

Self-Intensive Training on Full Stack Development

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Project Title: Bulk Mail Blocking/Unblocking

Domain: Students Affairs

Problem Statement:

Currently, the process of blocking or unblocking student email IDs for mass communication is inefficient and time-consuming. This is because it likely involves manual manipulation of email lists or contacting IT support for each individual student.

This inefficiency leads to several problems:

- **Delays in communication:** Important information or announcements may be delayed in reaching students due to the time required for manual blocking/unblocking.
- **Increased workload for IT staff:** IT staff may be burdened with repetitive tasks of managing student email access for bulk communication.
- **Potential for errors:** Manual processes are prone to human error, which could result in students being accidentally blocked or unblocked.

To address these issues, we propose the development of a secure and user-friendly portal system that allows authorized persons to efficiently block and unblock student email IDs for bulk communication purposes.

Project Description:

Build a portal system to block/unblock bulk bitsathy email ID of the students.

Project Requirements:

Here are some initial project requirements for developing a student email blocking/unblocking portal system:

Functional Requirements:

➤ **User Management:**

- ✓ The portal should allow for the management of user accounts with different access levels (e.g., administrator, students).

➤ **Student Management:**

- ✓ The system shall integrate with existing student databases to allow for easy selection of students by criteria (e.g., department, year).
- ✓ The system shall enable uploading of student lists in a standard format (e.g., CSV) for bulk blocking/unblocking.

➤ **Blocking/Unblocking Functionality:**

- ✓ The portal shall provide a user-friendly interface for selecting students and applying blocking/unblocking actions.
- ✓ The portal shall clearly indicate the current status (blocked/unblocked) of each student's email address.
- ✓ The portal should offer the ability to schedule blocking/unblocking actions for future dates.

➤ **Reporting and Auditing:**

- ✓ The system shall generate reports on blocking/unblocking activities, including timestamps, user who performed the action, and students affected.

Non-Functional Requirements:

➤ Security:

- ✓ The system shall be secure and compliant with relevant data privacy regulations.
- ✓ User access and student data must be protected with appropriate security measures.

➤ Performance:

- ✓ The system shall be responsive and handle bulk actions efficiently without significant delays.

➤ Scalability:

- ✓ The system should be scalable to accommodate a growing number of students and users.

➤ Usability:

- ✓ The system interface should be user-friendly and intuitive for authorized persons with varying levels of technical expertise.

➤ Additional Considerations:

- ✓ Integration with existing email infrastructure.
- ✓ User training materials on how to utilize the portal system effectively.
- ✓ Development of a disaster recovery plan to ensure system availability in case of outages.

This is a starting point, and the specific requirements may evolve as the project progresses and stakeholders are further consulted.

Technical Components:

Component	Tech Stack
Frontend	• HTML • CSS • JavaScript
Backend	• Linux • Apache Web Server • PHP with Laravel framework
Database	• MySQL
API	• REST Ful services

LAMP Stack:

The **LAMP stack (Linux, Apache, MySQL, PHP)** is a popular and powerful combination for developing web applications. Here's a breakdown of how you can build your **bulk mail blocking/unblocking portal** using LAMP:

➤ Setting Up the LAMP Stack:

- ✓ **Install LAMP:** I am preferring XAMPP Control Panel, Visual Studio code and MySQL workbench 8.0 CE.
- ✓ **Design:** Use MySQL to create a database to store candidate information. Design tables with relevant fields like Mail ID, Password, etc.

➤ Web Development with PHP:

- ✓ **Develop the Website Framework:** Use PHP to build the website's functionalities. Frameworks like Laravel or CodeIgniter can provide a foundation for user authentication, routing, and database interaction.

➤ **User Interface (UI) Design:**

- ✓ **HTML & CSS:** Use HTML to structure the website's content and CSS to style the visual elements.
- ✓ **User-Friendly Interface:** Design an intuitive interface for admin to easily navigate and manage candidate data. Consider using forms, tables, and clear labelling for data fields.

➤ **Security Considerations:**

- ✓ **User Authentication:** Implement a secure login system using PHP.
- ✓ **Regular Backups:** Implement a backup system for your database and website files to ensure data recovery in case of emergencies.

➤ **Deployment:**

- ✓ **Web Server Configuration:** Configure your web server (Apache) to serve the website files and interact with the MySQL database.
- ✓ **Testing & Deployment:** Thoroughly test the website functionalities before deploying it to a live environment. Consider user acceptance testing to ensure it meets your needs.

➤ **Limitations of LAMP Stack:**

While LAMP is a powerful choice, you might consider alternatives for specific needs:

- ✓ **Complexity:** Building a complex application from scratch requires significant development expertise.
- ✓ **Security:** Maintaining security requires ongoing vigilance and updates.

Workflow:

This workflow outlines the process for authorized personnel to utilize the student email blocking/unblocking portal system:

1. User Login:

- Authorized personnel access the portal using their username and password.

2. Student Selection:

- Users can choose individual students or select groups based on pre-defined criteria (e.g., department, year).
- Alternatively, users can upload a CSV file containing a list of student email addresses for bulk actions.

3. Action Selection:

- The user specifies whether to block or unblock the selected student email addresses.
- An option to schedule blocking/unblocking for a future date and time may be available.

4. Confirmation and Processing:

- The system displays a confirmation screen summarizing the selected students and the chosen action (block/unblock).
- Upon confirmation, the system executes the blocking/unblocking action for the designated students.

Additional Considerations:

- The system may provide functionalities for searching and filtering students to assist with specific selections.
- Users may be able to save frequently used student groupings for quicker selection in future tasks.
- The system should display clear notifications and informative messages throughout the workflow to guide users and confirm actions.

Flowchart:

